

Data_Analyst — Step 4

Learn Power BI & Tableau

Detailed Learning Notes • CareerCompass

1. Learning Objective

Build interactive dashboards and understand the data-preparation, modeling and visualization concepts behind BI tools.

2. Core Concepts — Detailed Breakdown

Power BI workflow

Connect to data, transform with Power Query, model relationships, create calculations with DAX, design report pages and publish securely.

Semantic modeling

Use a clear fact/dimension structure where appropriate, define relationships and avoid ambiguous filter paths. A correct model makes calculations more reliable.

DAX concepts

Understand measures versus calculated columns and why filter context matters. Start with simple aggregations before advanced expressions.

Tableau workflow

Connect to data, understand dimensions/measures, create worksheets, calculations, filters and dashboards.

Visualization design

Choose charts based on the question. Keep titles, units, filters and interaction behavior clear.

3. Recommended Learning Workflow

- Build a small Power BI report.
- Practice Power Query transformations.
- Create relationships and measures.
- Build the same question in Tableau.
- Compare usability and explain design choices.

5. Hands-on Project / Practice

Create a sales dashboard in Power BI with KPI cards, trend, category comparison, geography where relevant, filters and a drill-down. Recreate one key view in Tableau and write a short comparison of the two tools.

6. Common Mistakes & Pitfalls

- Building visuals before modeling data.
- Using calculated columns when a measure is more appropriate.
- Ambiguous relationships.

- Overusing slicers.
- Choosing flashy charts over interpretable ones.

7. Completion Checklist

- I can connect and clean data in Power BI.
- I understand relationships and basic DAX.
- I can create interactive report pages.
- I can build basic Tableau worksheets and dashboards.
- I can justify visualization choices.

8. Interview / Assessment Questions

- Measure vs calculated column?
- What is filter context?
- Why do relationships matter?
- When is Tableau useful?
- How do you choose a visualization?

9. Recommended References

Microsoft Power BI analytics: <https://learn.microsoft.com/en-us/training/paths/data-analytics-microsoft/>

Prepare and visualize with Power BI: <https://learn.microsoft.com/en-us/training/paths/prepare-visualize-data-power-bi/>

Model data with Power BI: <https://learn.microsoft.com/en-us/training/paths/model-power-bi/>

Tableau Free Training: <https://www.tableau.com/learn/training>

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.